

Factories and abstract factory

- Wholesale object creation (non-piecewise, like Builder) can be outsourced to
 - A separate method (factory method)
 - That may exist in separate class (factory)
 - Hierarchy of factory $\rightarrow$ abstract factories

factory method is typically any method which creates an object.

from math import *

class Point:

```
def __init__(self, x, y):  
    self.x = x  
    self.y = y
```

@staticmethod

```
def new_cartesian_point(x, y):  
    return Point(x, y)
```

@staticmethod

```
def new_polar_point(rho, theta):  
    return Point(rho * sin(theta), rho * cos(theta))
```

$\rightarrow$ factory methods

factory : following soc principle, we separate the factory method to a new entity. And since we are working in oop environment, we use class.

class PointFactory:

@staticmethod

def new-cartesian-point(x, y):
 return Point(x, y)

@staticmethod

def new-polar-point(rho, theta):
 return Point(rho * sin(theta), rho * cos(theta))